Regional Data and Digital Archives in the Age of AI

Searchable to Askable Archive

RAG-based Annals of the Joseon Dynasty AI chat-bot Architecture

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The slides and Ask Sillok

Slides

littlekg87.github.io/lecture/Asksillok_presentation

Ask Sillok

beta.asksillok.com

From sources you search
to sources you ask

Plain-language question

Did Taejo order walls to be built at Pyongyang?

Related article · Taejo 2, 1st month, 6th day

도당에 명하여 평양에 성을 쌓도록 하다

命都評議使司, 城平壤。

Original ↗

Back to keyword search ↶ Replay from the start ↶

Conceptual example of the difference between the two approaches

Local records, whose language opens them?

Researchers

“Gangneung · town wall · construction”

Precise searches using place names and specialist terms

The public

“Why was the wall in our town built?”

A clear question, but unfamiliar search terms

International visitors

“Why was the old wall in Gangneung built?”

Unfamiliar Korean place names and source language

Keyword-based search Useful when you know the vocabulary; harder when you are new to the archive.

Everyday and multilingual questions

AI connects the meaning of a question to places, topics and records

Local archive Documents · Photographs · Oral histories

Help people reach local records in their own words.

A proposed use for local archives. Questions are illustrative; multilingual retrieval needs validation.

RAG

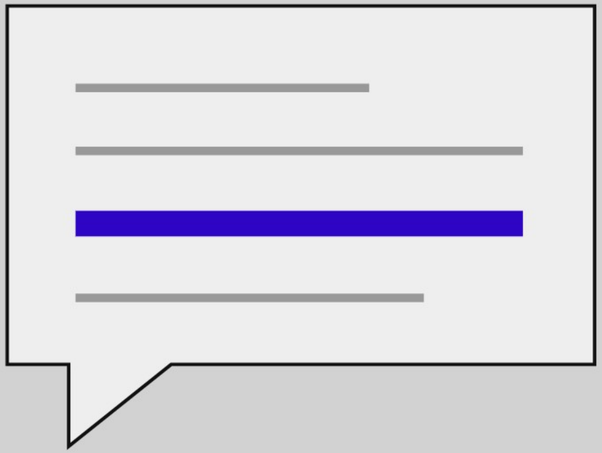
Retrieval-Augmented Generation

Generate answers grounded in specific evidence



Specific material

How the material is prepared +



An answer grounded in that material

How retrieval and answering work +

1

Building the DB

Turning sources into data you can ask

Start ↓

From records to a vector DB, design choices at every step.

A typical construction workflow



Chunking

→



Embedding

→



Vector DB construction

Collapse the Ask Sillok workflow ↑

01

Structure as XML

Preserve article structure,
then separate text and metadata

Link both archives with the same ID

02

Fine-tune the encoder

Learn the correspondence between
Korean titles and Classical Chinese text

Embed records with an adapted encoder

03

Store two vector versions

Base-model vectors
Fine-tuned-model vectors

Compare using the same chunk ID

Preserve source structure, train the encoder and compare both vector versions

An article in the Annals

Annals of the Joseon Dynasty Taejo Sillok

Taejo 2, 1st month, 6th day

도당에 명하여 평양에 성을 쌓도록 하다

○壬子/命都評議使司, 城平壤。

Ordered the Dopyeonguisasa to build walls at Pyongyang.

Class Military · Border defense (關防)

Reconstructed article view Open the actual article ↗ Convert to XML →

The Annals as XML data

ID | Metadata | Body | All

```
<level5 id="waa_10201006_001">
    <front><biblioData type="T">
        <mainTitle>도당에 명하여 평양에 성을 쌓도록 하다
        </mainTitle>
        <dateOccured date="1393-01-06L0" type="서기"/>
        <subjectClass>군사-관방(關防)</subjectClass>
    </biblioData></front>
    <text><content><paragraph>
        o壬子/命都評議使司, 城<index type="Place">平壤
        </index>。
    </paragraph></content></text>
</level5>
```

Selected part

One article

ID · metadata · body

Choose a part to see it highlighted.

← View as article | Actual XML excerpt · abridged | XML structure and source + | Separate into two archives →

Same ID: two archives joined

TEXT ARCHIVE

Body archive

ID waa_10201006_001

〇壬子/命都評議使司,
城平壤。

Text for training and retrieval

Same ID

METADATA ARCHIVE

Metadata archive

ID waa_10201006_001

Title	도당에 명하여 평양에 성을 쌓도록 하다
Reign · date	Taejo · 1393. 01. 06.
Subject	Military · Border defense
Source	Link to the original

Body ← same ID → Metadata Joined at search time, leading back to the original

Turning an article into numbers — Embedding

The article going into the model

waa_10201006_001

도당에 명하여 평양에 성을 쌓도록 하다

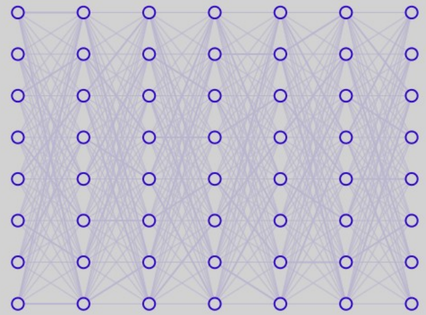
○壬子/命都評議使司, 城平壤。

Korean title + Classical Chinese body

→

Encoder

Embedding model (LLM)



Transformed by the trained weights

→

The article's vector

waa_10201006_001

1	-0.00061
2	-0.03906
3	-0.00476
4	0.02991
⋮	⋮



An array of 2,560 numbers

← Back to the archives

Excerpt of the actual stored vector · the process is a conceptual animation

Replay embedding ↻

Fine-tuning — training changes the weights

Forward pass → Compute loss ← Backpropagation Update weights

Training set

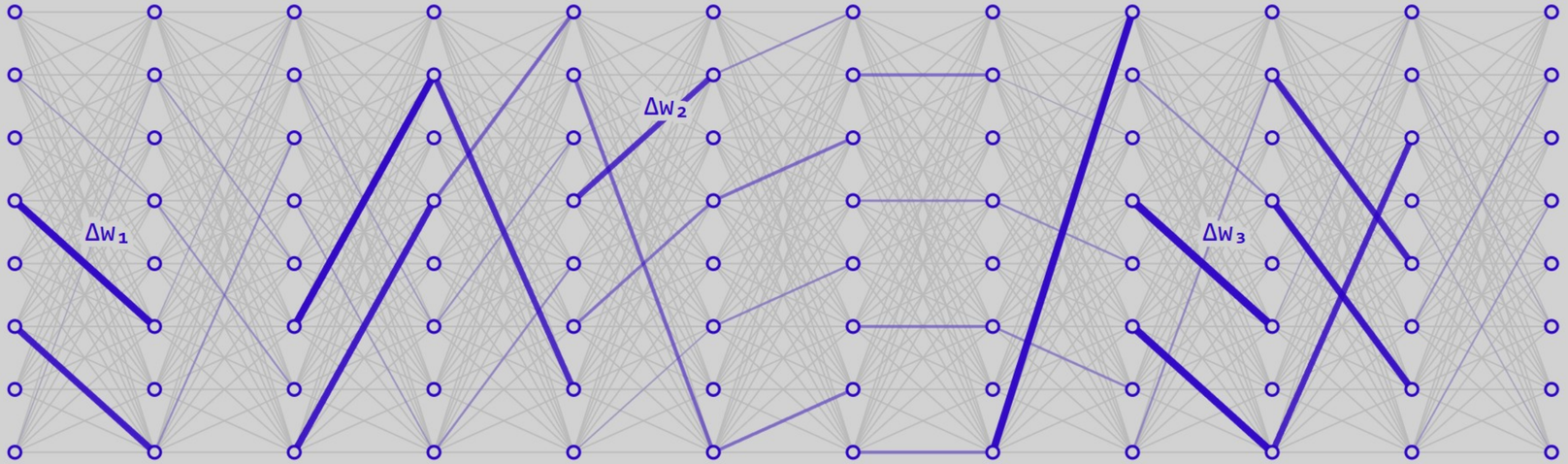
Korean title

도당에 명하여 평양에 성을 쌓도록 하다

↕ Same article

Classical Chinese body

命都評議使司, 城平壤。



Distance between embeddings



The two languages of one article are represented close together

Loss: the gap from the training target

Correction Δw₁ 0.0000 → -0.0271

Correction Δw₂ 0.0000 → +0.0203

Correction Δw₃ 0.0000 → -0.0297

Grey connections frozen · colored connections and numbers updated by training

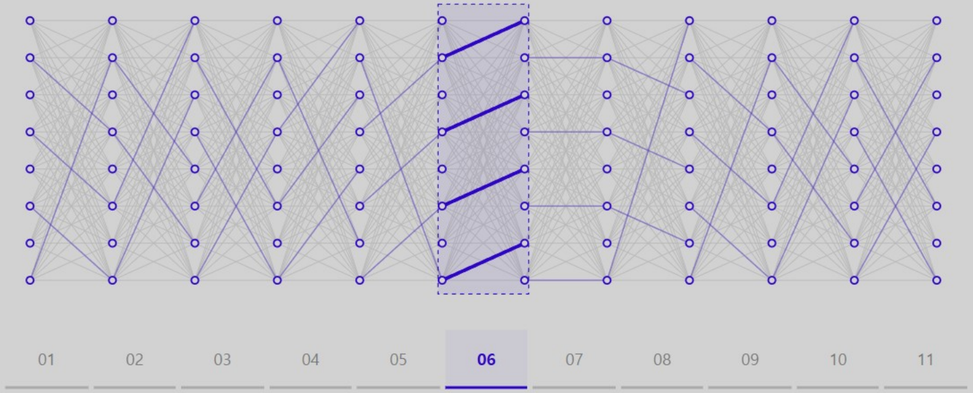
Compared after training | Back to before training ↶ | Round 3 / 3 done · corrections kept

Conceptual diagram of LoRA trained on Korean title–Classical Chinese body pairs · layer count, signals, corrections and distance changes are illustrative.

LoRA

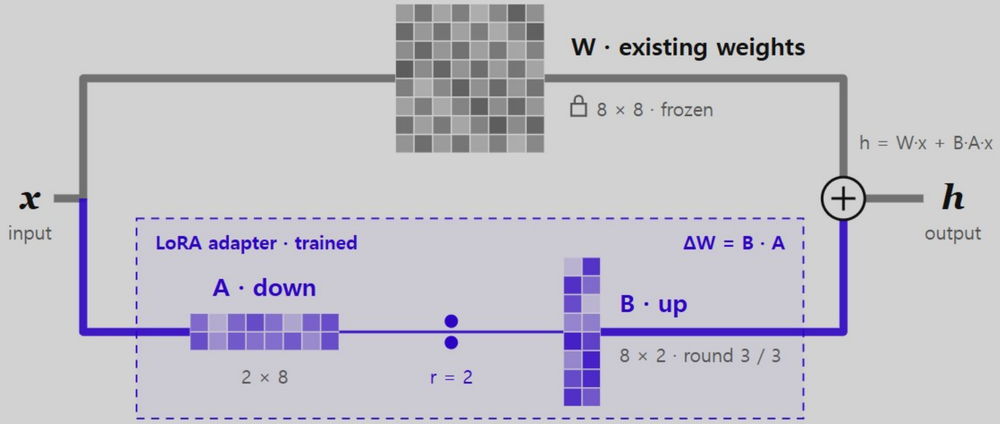
Freeze the existing weights; train only the adapter attached beside them

Same network, a different way of training



Choose a segment to zoom in on its W and adapter.

Selected segment 6



The 64 numbers of W stay as they are; **only the 32 numbers of A·B are trained.**

The gap grows with size. Instead of 1,024 × 1,024 = 1,048,576 numbers, r = 8 needs 16,384.

Full fine-tuning trains W itself

LoRA **W frozen · only adapter A·B trained (ΔW = B·A)**

More ↗

Train again ↻ Start over ↶

Training done · W unchanged, the output with the adapter added has changed

Same conceptual network as the fine-tuning scene · segments do not map one-to-one to Transformer layers · values and sizes (8 × 8, r = 2) are illustrative

This is how the Annals are stored

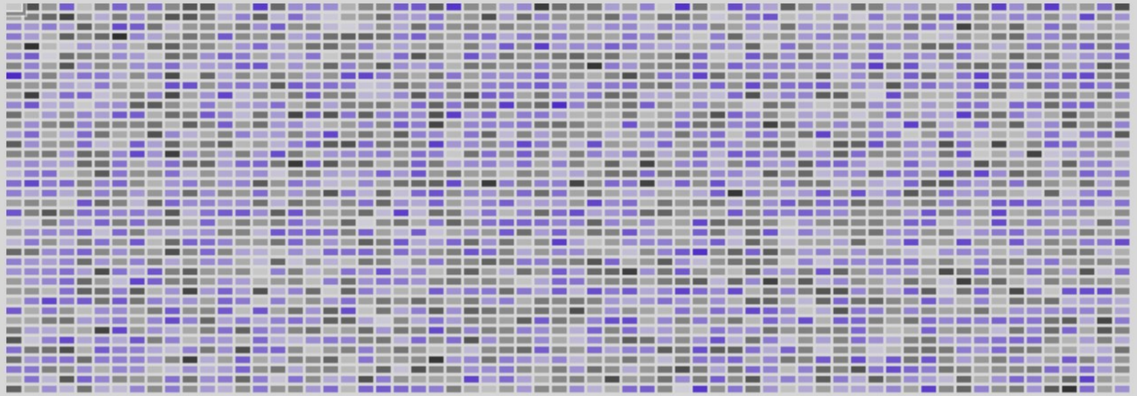
Maps for the new capital | Yeonboksa pagoda | **Pyongyang walls** | Court greeting | Stipends paid

도당에 명하여 평양에 성을 쌓도록 하다

2,560 dimensions · compared at the same positions and color scale

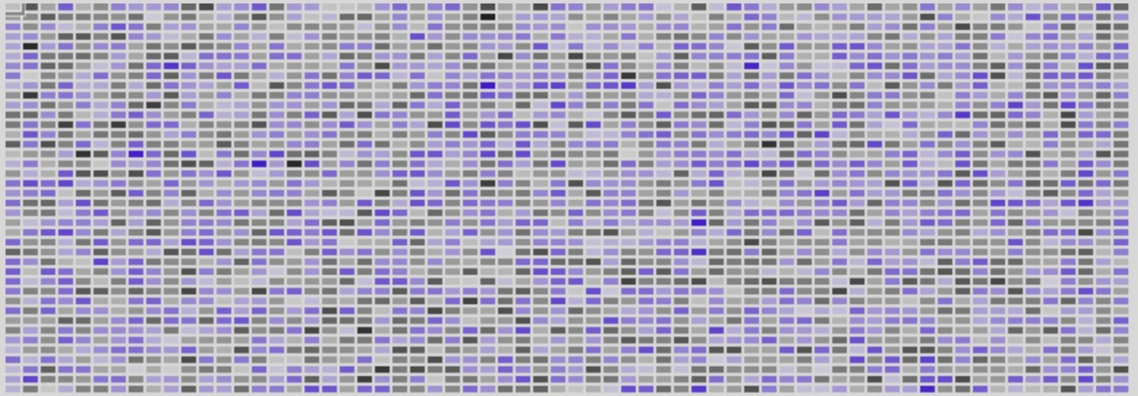
− 0 +

Base model



-0.0001373291015625

Fine-tuned model



-0.000606536865234375

Dimension 1 / 2,560

Five actual stored records · point at a cell to see both models' values · color differences do not indicate retrieval performance.

▶ View the original of the selected article

▶ Data and display

2

Retrieval

From a question to a grounded answer

Start ↓

From question to answer
through four carefully designed steps.

A typical retrieval and answer workflow



Question → **Vector DB search** → **Answer**

Collapse the Ask Sillok retrieval workflow ↑

01

Split the question

Query vector + metadata

02

Compare similarity and conditions

Vector similarity + condition match

03

Select candidates, add bonus

Nearest candidates → bonus for matching conditions

04

Answer with its source

Answer from evidence + link to the actual article

Split the question → compare → rank → answer with the evidence

Reranking by vector similarity + metadata match

Hybrid Retrieval · Re-ranking

STEP 01 — 03 · two routes of retrieval, one ranking

Split the question, compare, and rank

Question

Did Taejo order walls to be built at Pyongyang?

01 Process the question

Search the body data

Embed the whole question

Fine-tuned encoder → query vector

```
[0.02, -0.04, 0.13, 0.08, -0.11, 0.06, ...]
```

Metadata search

Extract only the conditions in the question

Person	Place	Subject
Taejo	平壤	Military · Border defense

02 Search each

Select candidates by vector similarity

Cosine similarity · higher is closer

A	Taejo · walls at Hanyang	0.92
B	Sejong · Pyongyang wall repair	0.90
C	Taejo · walls at Pyongyang	0.88

Top candidates by similarity: A · B · C

D (0.52) dropped

A bonus per item when a condition matches

Article	Person match +0.05 (Taejo)	Place match +0.05 (平壤)	Subject match +0.05 (Military · Border defense)
A	+0.05	—	+0.05
B	—	+0.05	+0.05
C	+0.05	+0.05	+0.05

Total metadata bonus

A	B	C
+0.10	+0.10	+0.15

Combine

03 Combine · final ranking

Vector similarity + Metadata bonus

			Vector		Bonus
1	C	1.03	0.88	+	0.15
2	A	1.02	0.92	+	0.10
3	B	1.00	0.90	+	0.10

C : 3rd → **1st**

All three conditions of the question match

Whole flow | Body search | Metadata search | Combine · final ranking

Articles, vectors and scores are illustrative · +0.05 each for person, place and subject matches (not production weights) · the combined score can exceed 1.

STEP 04 · generating the answer and linking its source

Every answer is linked to an actual article in the Annals

Question

Did Taejo order walls to be built at Pyongyang?

→

Evidence · the selected article

Taejo 2 · 1393, 1st month, 6th day

도당에 명하여 평양에 성을 쌓도록 하다

○壬子/命都評議使司, 城平壤。

→

LLM

Answer generated from the evidence

Model info ↗

→

Answer grounded in the source

Yes. According to the entry for the 6th day of the 1st month of Taejo 2 (1393), Taejo ordered the Dopyeonguisasa **to build walls at Pyongyang.** [1]

↖ Click the citation number for the actual article

Replay answer generation ↻ Start over ↻

Demo answer based on an actual Taejo Sillok article · not a live generated result.

Closing / Tech stack

This humanities project runs on NPU and AWS

NPU-accelerated embeddings · AWS-based retrieval pipeline

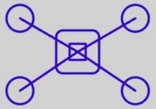


Research interface

React

Where the researcher's question comes in

→

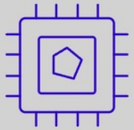


Cloud orchestration

FastAPI on AWS Lambda

A serverless API coordinates retrieval

→



Embedding compute

Qwen3 on NPU

Annals-specific embeddings

→



Vector search

AWS S3 Vectors

Storing and searching vectors

→



Evidence + answer

S3 Store + DeepSeek

Joining the originals, generating the answer

Embedding archive **Qwen3-Embedding-4B + LoRA** 2,560 dimensions · cosine

Production setup as in the earlier Ask Sillok presentation · model and dimensions from the archive manifest

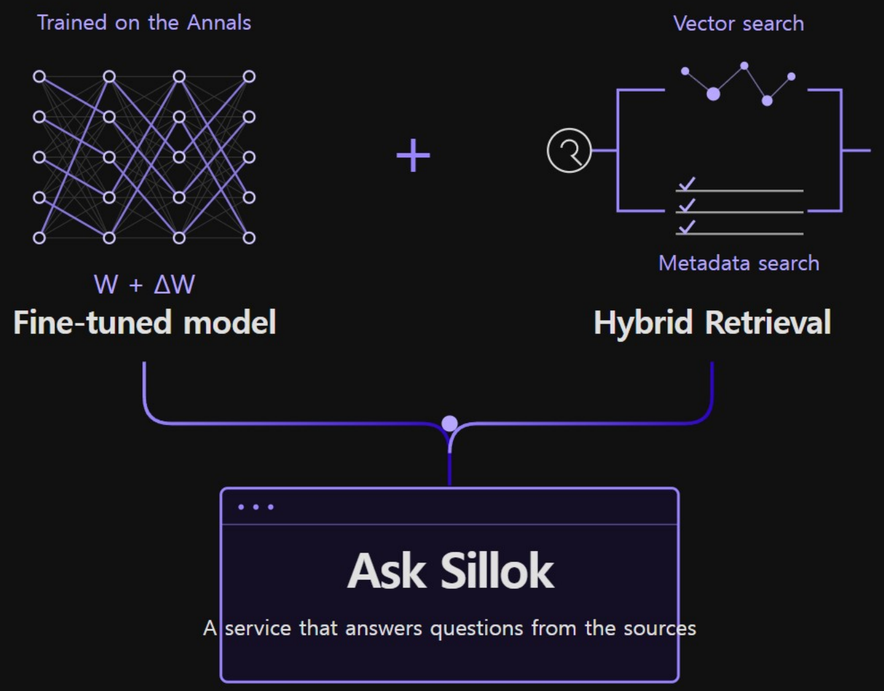
More ↗

Ask Sillok

Not an archive you search, an archive that answers

Open Ask Sillok ↗

External service · needs a network connection



Kunha Kim
Pukyong National University

Byungjun Kim
Academy of Korean Studies

Donghyun Woo
KAIST

Jeong-hui Park
Dongguk University

Si-heon Lee
Academy of Korean Studies

Ask Sillok research team

The service and this deck +

Local archives and AI

Local records become knowledge that anyone can ask about

Residents explore local memory. Students learn local history.
Visitors discover the stories of a place in their own language.

AI connects questions to records.
Archives ground answers in sources and context.

Thank you for listening

Slides

littlekg87.github.io/lecture/Asksillok_presentation

Ask Sillok

beta.asksillok.com

Portfolio

kunslab.com

Appendix

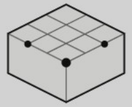
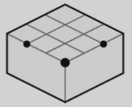
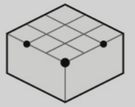
Model comparison and evaluation

Model comparison · expert evaluation · findings

Start ↓

Qwen3 Embedding · 0.6B / 4B / 8B

We fine-tuned three models ourselves and compared the performance of six.

Base model	 Qwen3 Embedding **0.6B** Base model	 Qwen3 Embedding **4B** Base model	 Qwen3 Embedding **8B** Base model
Fine-tuning	↓	↓	↓
Models we made	 Qwen3 Embedding **0.6B LoRA** Fine-tuned model ✓	 Qwen3 Embedding **4B LoRA** Fine-tuned model ✓	 Qwen3 Embedding **8B LoRA** Fine-tuned model ✓

Full comparison unfolded From the start ↺

We compared six models: three base and three we fine-tuned.

Models we fine-tuned **3** → 3 base + 3 fine-tuned **6** **models compared**

More ↗

EXPERT EVALUATION · BLIND

The relevance of the retrieved sources was judged by historians themselves.

01 Research questions →

Period-specific concepts | Modern vs. source vocabulary | Compound conditions | Proper nouns | Surface-match controls

What were the causes and results of the literati purges?

Questions designed by two historians of Joseon

02 Anonymized results →

Model names and distance scores hidden,

A | B | C | D

Model A · retrieved results ranking kept

01 →

02

03

⋮ top 20 per model

A–D assigned at random per rater and question

03 The expert's judgment

Model A · 1(rank)

Is this source relevant to the question?

0	1	2
Not relevant	Partly relevant	Clearly relevant

Demo choice: 2· not an actual expert rating

In the study, 10 raters each rated independently

50 Research questions × **4** Model × **20** Top sources × **10** Historians = **40,000** ratings

More ↗

Result lists and chosen scores only demonstrate the procedure · not actual results or expert ratings · IJKH manuscript §3.2

Findings and implications

Better machine performance does not guarantee usefulness for research.

Benchmark evaluation

Performance improved.

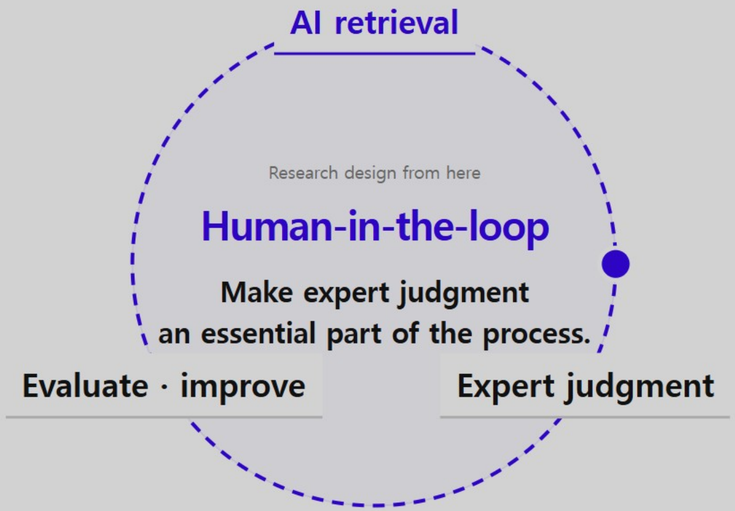
Finding the paired body from a title got better

≠

Expert evaluation

The gains were uneven.

Effects differed by model and question type,
and in some cases retrieval got worse.



AI finds candidate sources across a wide body of material.

AI is not a cure-all. **Technical progress and expert judgment have to go together.**

Summary of the results and discussion in the IJKH submission · the cycle below is the direction for future research design

Related work

1. Kunha Kim, “How Data Changes AI Answers: A Comparative Experiment on RAG Responses by Data Structuring of the Sejong Sillok Jiriji” (in Korean), Hanmunhak Nonjip 72, 2026.

2. Kunha Kim, Byungjun Kim, Donghyun Woo, Jeong-hui Park, Si-heon Lee, “Fine-Tuned Embeddings in Historical RAG: Benchmark Performance and Expert Evaluation on the Veritable Records of the Chosŏn Dynasty”, under review.

3. Kunha Kim and 4 others, “System, Search Method and Index Construction Method for Cross-Lingual Retrieval of Chronological Historical Sources”, Korean patent application, 2025.08.26.

4. Kunha Kim and 4 others, “Method and Apparatus for Training a Cross-Lingual Encoder Based on Title–Body Correspondence in Chronological Historical Sources”, Korean patent application, 2025.08.26.